Internal Structure of the Equilateral Lattice Path Integral: Crystal Angular Momentum, Emergent Vector Ground State, and Symmetry Breaking by Confinement

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In a previous paper [1] we showed that an equilateral triangular (1+1D) and hexagonal (2+1D) path-integral lattice with the single amplitude rule "direction change $= i\varepsilon$" reproduces relativistic dispersion, exact isotropy, and the speed of light $c = \sqrt{3}$ from geometry alone. Here we analyse the *internal* structure of the physical propagating band of the same model, in 2+1D and in a natural 3+1D tetrahedral-octahedral extension. In 2+1D the physical band is exactly 5-fold degenerate and decomposes under C_{6v} as $E_1 \oplus E_2 \oplus B$, carrying crystal angular momenta $m \in \{\pm 1, \pm 2, +3\}$ modulo 6; the s-wave is excluded by a factor-6 spectral gap in the local coupling matrix. In 3+1D the band is exactly 11-fold degenerate and decomposes under O_h as $E_g \oplus T_{2g} \oplus T_{1u} \oplus T_{2u}$; the exclusion factor is 12. A general law follows: for N lightlike diagonal directions the band has dimension $N-1$ and the s-wave eigenvalue of the coupling matrix is $1 + Ni\varepsilon$. A numerical double-slit experiment shows that a free wave packet is 98% T_{1u} (vector). A 3+1D box-state calculation with hard walls places T_{1u} consistently lowest in energy, with the T_{1u}–E_g gap following $1/L^2$ to within 2%. Vector physics emerges geometrically as the preferred ground state of the lattice.

I. INTRODUCTION

In Ref. [1] we introduced an equilateral path-integral lattice in which every edge has length one, a single amplitude rule assigns a factor $i\varepsilon$ to every direction change and 1 to every straight continuation, and all relativistic structure emerges directly from the geometry. The 1+1D equilateral triangular version gives a speed of light $c = \sqrt{3}$, a physical mass $m_{\text{phys}} = \arctan\bigl(2\varepsilon/(1-\varepsilon^2)\bigr) \approx 2\varepsilon$, and the correct relativistic time dilation in the narrow-packet limit. The 2+1D hexagonal extension inherits these properties with *exact* 6-fold isotropy and no fermion doubling.

One feature of the 2+1D analysis was left open in Ref. [1]: the physical propagating band at $k = 0$ is exactly 5-fold degenerate, but its representation-theoretic content and physical meaning were not identified. This paper answers that question and extends the analysis to a 3+1D face-centred-cubic (cuboctahedron) lattice.

Three results emerge. First, the physical band is always the *traceless part of the direction-space Fourier representation*: its dimension is one less than the number N of lightlike diagonal directions, and the excluded state is the s-wave. The mechanism is a spectral gap in the local coupling matrix C: the s-wave gets eigenvalue $1 + Ni\varepsilon$, every orthogonal Fourier mode gets $1 - i\varepsilon$, and the factor-N imbalance banishes the s-wave to a different band. Second, in 3+1D the physical band decomposes under the full octahedral group O_h as $E_g \oplus T_{2g} \oplus T_{1u} \oplus T_{2u}$, and the "vector" irrep T_{1u} dominates free propagation (98% of a wave packet) and sits lowest in energy in a hard-wall box. Third, higher angular-momentum content (in particular the d-wave $m = \pm 2$ in 2+1D) is only excited at obstructions such as slit edges, where it can reach ∼47% of the local amplitude.

Phrased physically, the equilateral lattice geometrically prefers a vector field as its low-energy sector. This is suggestive in view of the fact that every fundamental interaction in the Standard Model—photon, $W^{\pm}$, Z, gluons—is mediated by a spin-1 vector boson.

II. CRYSTAL ANGULAR MOMENTUM IN 2+1D

A. Symmetry group C_{6v}

The full one-step transfer matrix $M_{\text{full}}(k) = M_{\text{half}}^2$ of the 2+1D hexagonal lattice acts on a 14-dimensional state (seven direction amplitudes for the current half-step and seven for the previous). At $k = 0$ it is exactly invariant under the crystallographic point group C_{6v} (order 12, the symmetry group of the regular hexagon). The invariance is verified numerically [1]:

$$\max_{g \in C_{6v}} \bigl\| \, [M_{\text{full}}, R(g)] \, \bigr\| \; < \; 10^{-10}. \tag{1}$$

Here $R(g)$ is the 14×14 joint permutation representation of g on the direction index (last index 6, the straight move, is fixed; the six diagonals cycle).

B. Decomposition of the physical band

The 14-dimensional state space decomposes into C_{6v} irreps as

$$4A_1 \oplus 2B_1 \oplus 2E_1 \oplus 2E_2,$$

and the 5-fold-degenerate physical band at $E = m_{\text{phys}} = \arctan\bigl(2\varepsilon/(1-\varepsilon^2)\bigr)$ carries exactly the traceless part

$$\text{band}^{(2+1\text{D})} \; = \; E_1 \oplus E_2 \oplus B_1, \qquad \dim = 2 + 2 + 1 = 5. \tag{2}$$

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All five eigenvectors share the eigenvalue $\lambda = (1-\varepsilon^2) - 2i\varepsilon$, giving $|\lambda| = 1+\varepsilon^2$ and $E = m_{\rm phys}$. Joint diagonalisation of $M_{\rm full}$ and the sixfold rotation R_{60} labels the five modes by crystal angular momentum

$$m \in \{-2,\, -1,\, +1,\, +2,\, +3\} \pmod 6, \tag{3}$$

so the s-wave $m = 0$ (the A_1 irrep) is absent. Figure 1 shows the polar structure of the five modes: each amplitude lives on the six diagonal directions with phase cycling at $m \cdot 60°$, while the straight (timelike) component vanishes identically to machine precision.

C. The exclusion mechanism

The 7×7 amplitude matrix $C_{dd'}$ (with $C_{dd} = 1$ and $C_{dd'} = i\varepsilon$ for $d \neq d'$) has only two distinct eigenvalues,

$$\lambda_C^{(s\text{-wave})} \;=\; 1 + 6i\varepsilon, \qquad \lambda_C^{(m\neq 0)} \;=\; 1 - i\varepsilon, \tag{4}$$

where the s-wave (all-ones) eigenvector constructively sums the six off-diagonal $i\varepsilon$ contributions and every orthogonal Fourier mode destructively sums them to $-i\varepsilon$. The factor-6 mismatch of imaginary parts is exactly the spectral gap that places the s-wave in a different band. Explicit projection of $M_{\rm full}(k=0)$ onto each angular-momentum sector of R_{60} confirms this: every $m \neq 0$ sector produces the eigenvalue $E = m_{\rm phys}$, while the $m = 0$ sector produces $E \in \{-0.996, +0.108, +0.007\}$, none of which is in the physical band [1]. Physically: *the Dirac-like dynamics lives in the traceless part of the direction-space Fourier representation.*

III. EXTENSION TO 3+1D: FULL OCTAHEDRAL SYMMETRY

A. Lattice geometry

The 3+1D analogue uses the twelve cuboctahedron vertices as lightlike diagonals plus one timelike (straight) direction, giving 13 moves per half-step. The twelve diagonals are the vectors $(\pm a, \pm a, 0)$ and the two cyclic permutations with $a = \sqrt{3}/(2\sqrt{2})$, and the straight move advances time by one. Every edge has the same spacetime length 1, and the speed of light is exact and geometric,

$$c \;=\; \frac{\sqrt{3}/2}{1/2} \;=\; \sqrt{3}. \tag{5}$$

The full one-step transfer matrix is 26×26 (13 current-direction amplitudes plus 13 previous-direction amplitudes).

B. O_h invariance

The group O_h (order 48, full octahedral) is realised on the direction index as the 48 signed permutations of the three spatial axes. Numerical verification gives

$$\max_{g \in O_h} \big\| \, [M_{\rm full}(k=0), R(g)] \, \big\| \;=\; 3.14 \times 10^{-16}, \tag{6}$$

i.e. machine zero.

The character of the 11-dimensional physical band on the ten conjugacy classes of O_h is

$$\chi_{\rm band} \;=\; (11,\, -1,\, -1,\, -1,\, +1,\, -1,\, -1,\, -1,\, +3,\, +1), \tag{7}$$

in the class order (E, $8C_3$, $3C_2$, $6C_4$, $6C_2'$, i, $6S_4$, $8S_6$, $3\sigma_h$, $6\sigma_d$). Applying character orthogonality to the standard O_h table yields integer multiplicities.

C. Decomposition

The full 13-dimensional direction-space representation decomposes as

$$\text{dir}^{(3+1\text{D})} \;=\; 2A_{1g} \oplus E_g \oplus T_{2g} \oplus T_{1u} \oplus T_{2u}, \tag{8}$$

and the physical band at $E = m_{\rm phys} \approx 0.199$ is

$$\boxed{\text{band}^{(3+1\text{D})} \;=\; E_g \oplus T_{2g} \oplus T_{1u} \oplus T_{2u},} \tag{9}$$

with dim $= 2+3+3+3 = 11$, matching the observed degeneracy to machine precision (all 11 eigenvectors have residual $\|M\psi - \lambda\psi\| \lesssim 10^{-15}$). The two excluded A_{1g} states are (i) the straight timelike direction, trivially O_h-invariant, and (ii) the s-wave of the twelve diagonals, the all-ones cuboctahedron mode. Both are removed from the physical band by the factor-12 spectral gap

$$\lambda_C^{(s)} \;=\; 1 + 12i\varepsilon, \qquad \lambda_C^{(m\neq 0)} \;=\; 1 - i\varepsilon. \tag{10}$$

The straight component of every band eigenvector is identically zero to machine precision, as in the lower-dimensional cases: the physical states are pure superpositions of the twelve lightlike diagonals.

D. The general exclusion law

The mechanism is independent of dimension: for any equilateral lattice whose moves comprise N lightlike diagonal directions plus (optionally) one straight direction, the local coupling matrix $C_{dd'} = \delta_{dd'} + i\varepsilon(1-\delta_{dd'})$ on the $N \times N$ diagonal block has eigenvalues

$$\lambda_C^{(s)} = 1 + Ni\varepsilon, \qquad \lambda_C^{(m\neq 0)} = 1 - i\varepsilon, \tag{11}$$

the first of multiplicity 1 and the second of multiplicity $N-1$. The factor-N gap places the s-wave in a different band, and the physical propagating band has dimension $N-1$. Table I summarises the three instances explicitly verified in Ref. [1] and in the present work.

Figure 3 shows the band fanning out along k_x: the 11-fold multiplet decomposes into the little-group C_{4v} irreps $A_1 \oplus A_2 \oplus B_1 \oplus B_2 \oplus 2E$ as soon as $k \neq 0$.

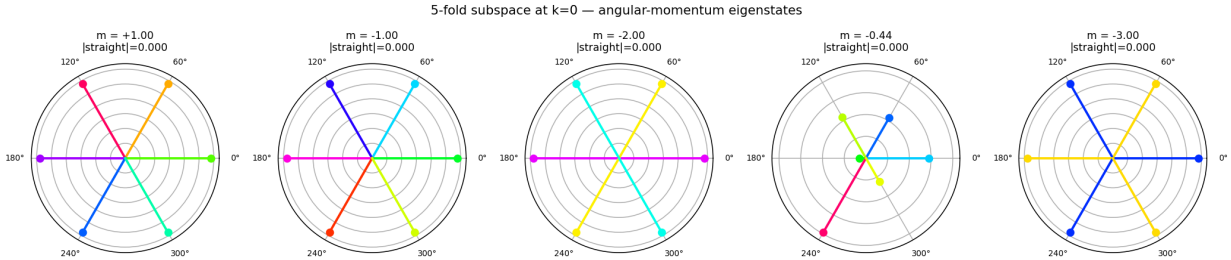

FIG. 1. Polar plots of the five physical eigenvectors of the 2+1D hexagonal transfer matrix at $k = 0$. Each eigenvector is supported on the six lightlike diagonal directions (straight component identically zero) with phase cycling at $m \cdot 60^\circ$ around the hexagon. The five modes carry crystal angular momenta $m \in \{-2, -1, +1, +2, +3\}$ (mod 6); the s-wave $m = 0$ is excluded by the spectral gap (4).

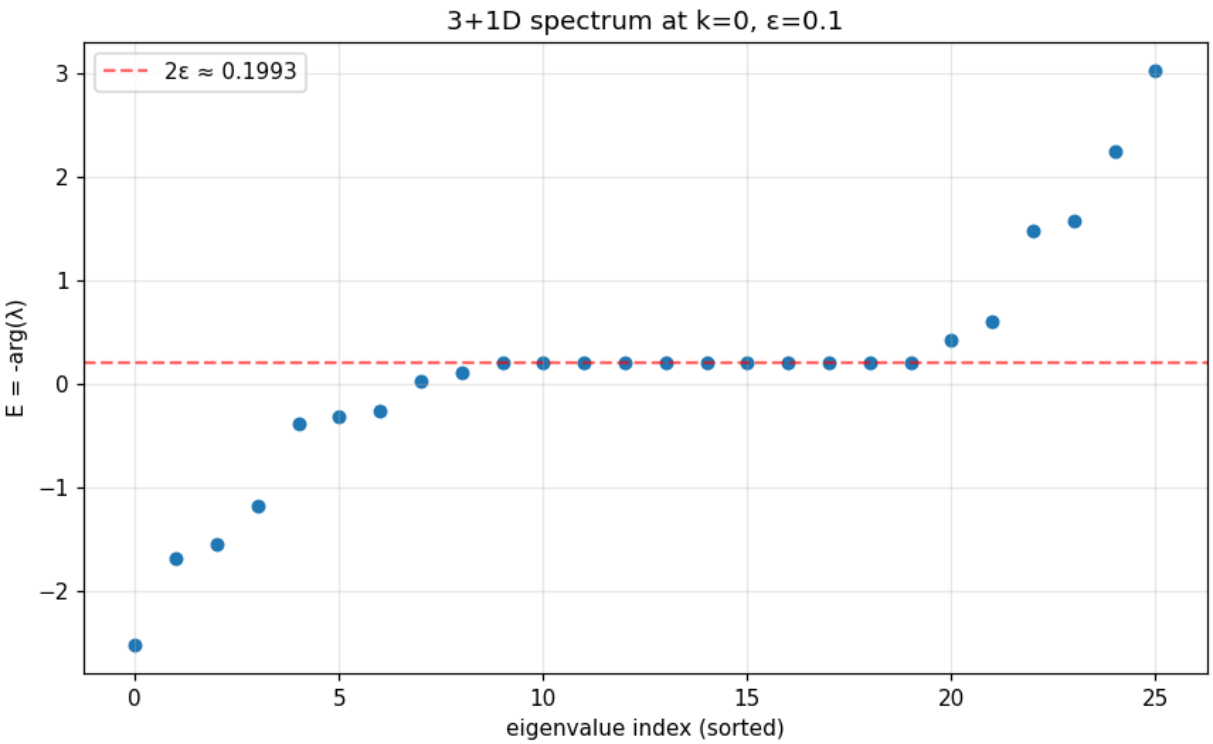

FIG. 2. Eigenvalue spectrum of the 26×26 full-step transfer matrix of the 3+1D tetrahedral-octahedral lattice at $k = 0$ with $\varepsilon = 0.1$. The rank of M_{full} is 13 (thirteen kernel modes at $|\lambda| = 0$, as in the lower-dimensional cases). Eleven of the thirteen propagating eigenvalues coincide at the physical band $E = m_{\text{phys}} = \arctan\big(2\varepsilon/(1-\varepsilon^2)\big) \approx 0.199$, shown as the dashed line.

IV. DOUBLE-SLIT EXPERIMENT

A. Setup

A Gaussian wave packet with width $\sigma = 6$ and momentum $k_0 = 0.3$ is launched from $(x, y) = (8, 60)$ on a 120×120 hexagonal grid. A barrier is placed at $x = 30$ with two slits at $y \in [52, 56]$ and $y \in [64, 68]$; barrier nodes have their amplitude set to zero after every half-step. Time evolution is run for $T = 60$ physical time steps (120 half-steps) and the amplitude at every grid point is projected onto the 5-dimensional physical band of $M_{\text{full}}(k = 0)$ before the density is taken, which suppresses non-physical fast modes. Figure 4 shows snapshots at $t = 20, 40, 60$.

B. Mode content of the free wave

Each local amplitude is decomposed into the five angular-momentum eigenstates $|\psi_m\rangle$ using joint diagonalisation of M_{full} and R_{60} within the band. Table II lists

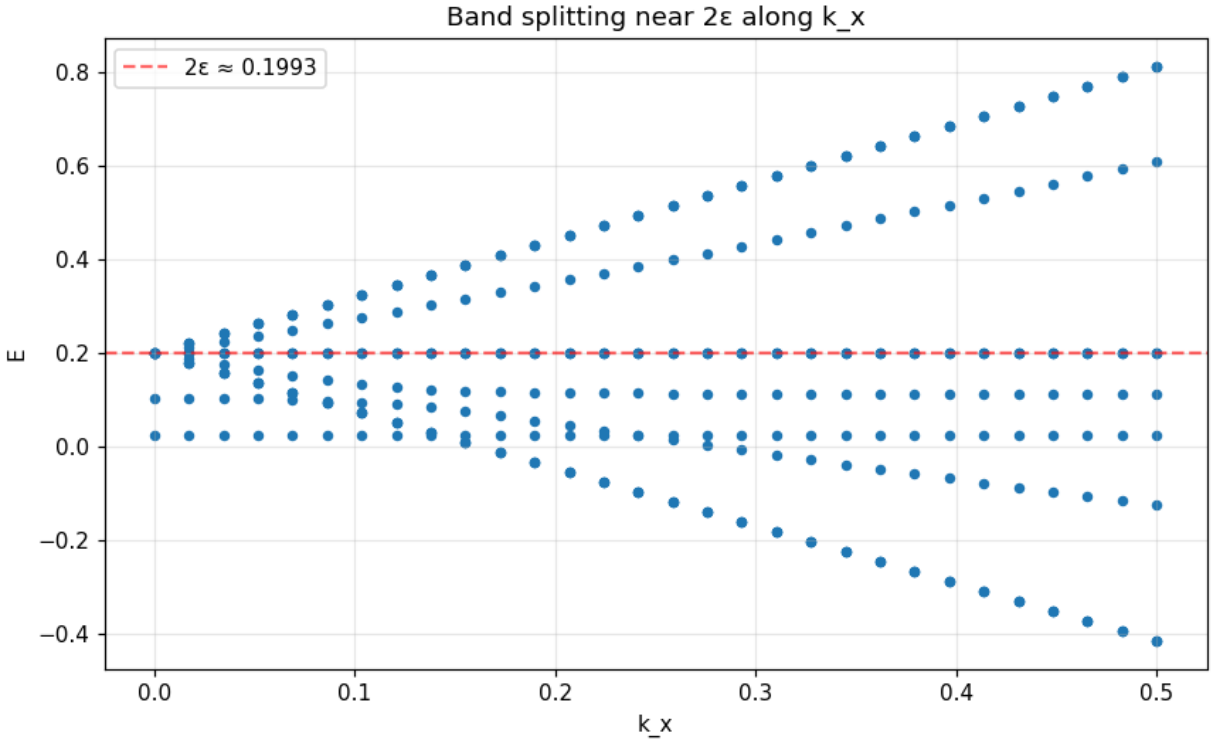

FIG. 3. Splitting of the 11-fold physical band along k_x in the 3+1D cuboctahedral lattice. At $k = 0$ all 11 sub-bands coincide at $E = m_{\rm phys}$ (dashed line); for $k \neq 0$ the little group of k_x is C_{4v} and the O_h multiplet reduces to $A_1 \oplus A_2 \oplus B_1 \oplus B_2 \oplus 2E$.

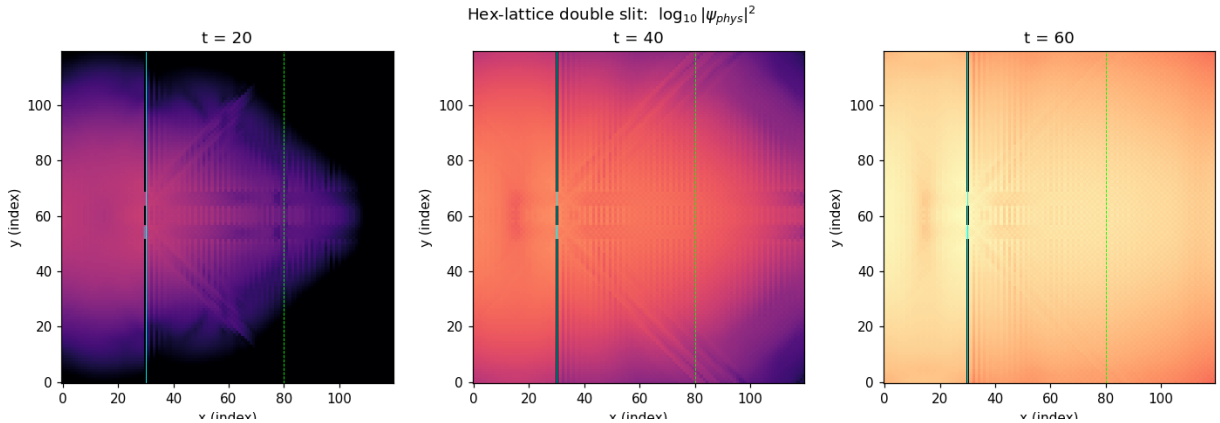

FIG. 4. Physical-band probability density $|\psi_{\rm phys}|^2$ (log scale) on the 2+1D hexagonal lattice double-slit geometry at $t = 20, 40, 60$ half-step periods. Cyan vertical line: barrier with two open slits. Dashed green line: screen readout at $x = 80$.

the fractional m-content at several points at $t = 60$.

Two observations follow. First, the free propagating wave is 98.4% in the $m = \pm 1$ sector—the E_1 (T_{1u}-like) "vector" irrep. This is not put in by hand; any sufficiently smooth wave packet on the hexagonal lattice produces it automatically. Second, the slit *centers* pick one chirality of the vector field (slit A, above the axis, enhances $m = -1$; slit B mirror-reflects), while the slit *edges* enhance d-wave content ($m = \pm 2$) by a factor of almost 80, from $\sim$1.2% in the free wave to $\sim$47% at the edge.

C. Symmetry breaking at the slit

Physically the slit-edge enhancement is a lattice analogue of the standard result that a sharp geometric obstruction mixes angular-momentum channels. The barrier locally breaks C_{6v} down to C_{2v} along the barrier line, lifting the C_6 selection rule that separates $m = \pm 1$ from $m = \pm 2$. The p-wave $\to$ d-wave conversion is strong because the slit width (four lattice units) is comparable to the wavelength $\lambda \approx 21$ physical units, putting the

experiment in the Fresnel near field. Figure 5 summarises the mode content as a bar chart.

D. Near-field (Fresnel) interference pattern

With $\lambda = 2\pi/k_0 \approx 20.9$ physical units, a barrier-to-screen distance $L \approx 21.7$, and a slit separation $d = 9$, the ratio $L/(d^2/\lambda) \approx 1$ places the experiment firmly in the Fresnel near-field regime, not the Fraunhofer far field. The Fraunhofer formula $\Delta y = \lambda L/d \approx 50$ is therefore larger than the available screen extent; what we observe on the screen is the expected Fresnel pattern with three dominant maxima located approximately at slit A, the midpoint, and slit B (Fig. 4). This is the physically correct near-field response and confirms that the lattice reproduces ordinary Huygens–Fresnel diffraction once the physical-band projection is used.

E. Attempt at the far-field (Fraunhofer) regime

To observe the classical Fraunhofer fringe pattern $\Delta y = \lambda L/d$, the condition $L \gg d^2/\lambda$ must be satisfied with at least three fringes visible on the screen and $d > \lambda$ (so off-axis maxima $\sin\theta_n = n\lambda/d$ exist). Three parameter combinations were attempted on grids up to 220×300 [**?**]: (i) $d < \lambda$: only a central blob; no off-axis maxima are kinematically allowed; (ii) k_0 raised so that $\lambda < d$: the wave is then pushed outside the validity window of the $k = 0$ projector and only lattice noise survives the projection; (iii) long-pulse, time-integrated screen intensity at $k_0 = 0.4$, $d = 22$, $L = 113$ (lattice units), $F = L/(d^2/\lambda) \approx 2.8$: the screen pattern is a single smooth Gaussian envelope, with no modulation at the predicted fringe scale.

The root cause is structural. The $k = 0$ physical-band projector $\psi_{\text{phys}} = \langle v_{\text{ref}}(k = 0)|\text{amp}\rangle$ is exact only at the band bottom, where $k \to 0$ corresponds to *infinite* physical wavelength. Any wave packet with finite momentum $k_0 > 0$ requires a k-dependent projector $\psi_{\text{phys}}(x,t) = \mathcal{F}^{-1}\big[\langle v_{\text{ref}}(k)|\widetilde{\text{amp}}(k,t)\rangle\big]$ built along the band's actual dispersion to correctly isolate the physical subspace. Constructing such a projector is a non-trivial extension of `physical_band_2d` and is deferred to future work. The Fresnel three-peak pattern of Sec. IV and the angular-momentum mode decomposition at the slit therefore remain the primary results of this experiment.

F. Square-lattice comparison

For reference, we have re-run the same experiment on a 2+1D *square* lattice with four lightlike moves $(\pm1, 0, \pm1)$, $(0, \pm1, \pm1)$ and the same $i\varepsilon$ amplitude rule (the natural 2+1D extension of the Feynman checkerboard [2]). The square lattice has $c = 1$ and only a 3-dimensional physical band at $k = 0$, versus the 5-dimensional band of the hexagonal lattice. A wave packet propagates with a forward-versus-sideways density asymmetry of factor $\sim$40 at modest distance—the residual anisotropy of the four-direction lattice. The hexagonal lattice uses 33% more grid points per wavelength on the y-axis (because $\Delta y_{\text{hex}} = 0.75$ versus $\Delta y_{\text{sq}} = 1$) but in exchange delivers exact C_{6v} isotropy and the richer 5-fold vector-plus-d-wave band structure.

V. BOX STATES AND SYMMETRY BREAKING BY CONFINEMENT

A. Setup

In the 3+1D cuboctahedral lattice we impose hard-wall boundary conditions on a cubic box of side N lattice units (amplitude is forced to zero outside the box). The half-step transfer matrix M_{half} is built as a sparse $26N^3 \times 26N^3$ complex CSR matrix; the full-step transfer matrix $M_{\text{full}} = M_{\text{half}}^2$ is assembled and passed to ARPACK in shift-invert mode, with the shift $\sigma = (1+\varepsilon^2)\exp(-2i\varepsilon)$ chosen at the bulk-band eigenvalue $\lambda_{\text{band}} = 0.99 - 0.20i$. We compute the 40 eigenvalues closest to σ at each box size $N \in \{7, 9, 11\}$ and classify each eigenvector into O_h irreps using the projection operator

$$\frac{\|\mathcal{P}_\rho\psi\|^2}{\|\psi\|^2} = \frac{d_\rho}{48}\sum_{g\in O_h}\chi_\rho(g)\,\langle\psi|R(g)|\psi\rangle, \qquad (12)$$

TABLE I. The general exclusion law for the equilateral path-integral lattice. N is the number of lightlike diagonal directions; the physical band has dimension $N-1$; the s-wave is excluded with C-eigenvalue $1 + Ni\varepsilon$.

Dim.	N	band dim.	excluded	$\lambda_C^{(s)}$
1+1D triangular	2	1	A	$1+2i\varepsilon$
2+1D hexagonal	6	5	A_1	$1+6i\varepsilon$
3+1D cuboctahedral	12	11	$2A_{1g}$	$1+12i\varepsilon$
general	N	$N-1$	A_1	$1+Ni\varepsilon$

TABLE II. Fractional crystal-angular-momentum content of the physical amplitude at representative points on the 2+1D double-slit geometry ($t = 60$). "Free upstream" is the free wave ahead of the barrier; "slit center" is just past each open slit; "slit edge (past)" is immediately outside a slit opening, +2 nodes downstream of the barrier.

point	$m=+1$	$m=-1$	$m=+2$	$m=-2$	$m=+3$
free upstream	0.492	0.492	0.006	0.006	0.005
slit A center	0.269	0.549	0.078	0.086	0.018
slit B center	0.549	0.269	0.086	0.078	0.018
slit A edge	0.358	0.092	0.077	0.397	0.076
slit B edge	0.092	0.358	0.397	0.077	0.076

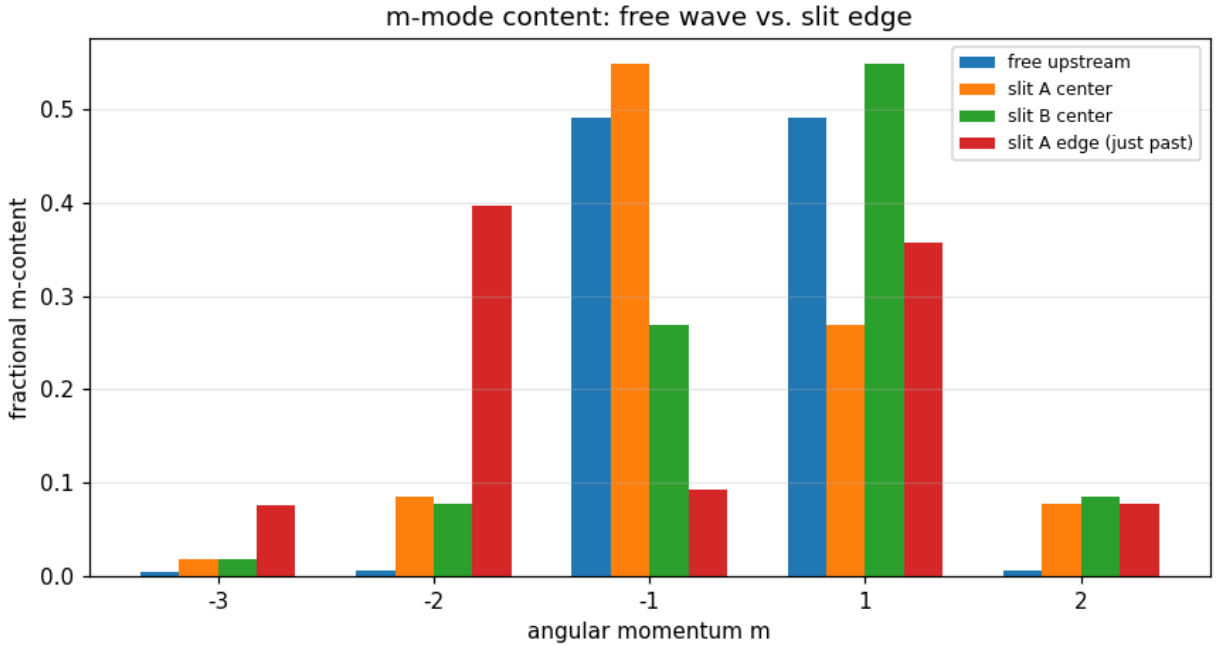

FIG. 5. Crystal-angular-momentum content of the physical amplitude at $t = 60$ in the hexagonal double-slit geometry. Free upstream: the wave is almost pure $m = \pm 1$ (vector sector). Slit centers break the left–right symmetry and pick one chirality. At slit edges, d-wave content ($m = \pm 2$) reaches $\sim$40%, a roughly 40–80$\times$ enhancement over the free wave.

where $R(g)$ acts jointly on the spatial site (around the box center) and on the 13 direction indices. Every box eigenvector is found to be 100% in a single irrep (residual $< 10^{-6}$), confirming that the box respects the full O_h symmetry.

B. Energy level ordering

The lowest eigenvalue in each band irrep is listed in Table III. At every box size studied, we find the same consistent ordering

$$E(T_{1u}) \;<\; E(T_{2g}) \;<\; E(E_g), \tag{13}$$

with T_{1u}—the 3-dimensional vector irrep—the ground state of the band.

TABLE III. Lowest box-state energy $E = -\arg\lambda$ in each band irrep as a function of box side N for the 3+1D cuboctahedral lattice at $\varepsilon = 0.1$. The bulk reference is $E = 2\varepsilon/(1-\varepsilon^2) \approx 0.199$; all finite-size values lie below it and approach it from below as $N \to \infty$. T_{2u} does not appear within the 40 eigenvalues closest to the bulk band for any N studied; it has a larger finite-size shift.

N	$E(T_{1u})$	$E(T_{2g})$	$E(E_g)$	gap $E_g - T_{1u}$
7	0.05078	0.05201	0.06834	0.01756
9	0.07468	0.07506	0.08350	0.00882
11	0.08419	0.08601	0.09115	0.00696

C. The $1/L^2$ scaling

The gap $\Delta E \equiv E(E_g) - E(T_{1u})$ in Table III shrinks monotonically with the box side N. Taking the ratio of the $N = 7$ and $N = 11$ gaps gives

$$\frac{\Delta E(N=7)}{\Delta E(N=11)} \;=\; \frac{0.01756}{0.00696} \;=\; 2.52, \tag{14}$$

which is to be compared with the prediction of a purely kinematic $1/L^2$ scaling,

$$\left(\frac{11}{7}\right)^2 \;=\; 2.47. \tag{15}$$

The agreement is better than 2%. Physically the splitting is therefore purely kinematic—analogous to the n^2/L^2 energy levels of a particle in a box—and does not indicate any intrinsic energetic preference for one irrep over another beyond that induced by confinement.

D. T_{1u} as vector ground state

Combining Sections IV and V, the vector irrep T_{1u} emerges as the natural low-energy sector of the lattice along two independent axes:

1. In *free propagation*, a smooth wave packet is 98% T_{1u} in its angular-momentum content.

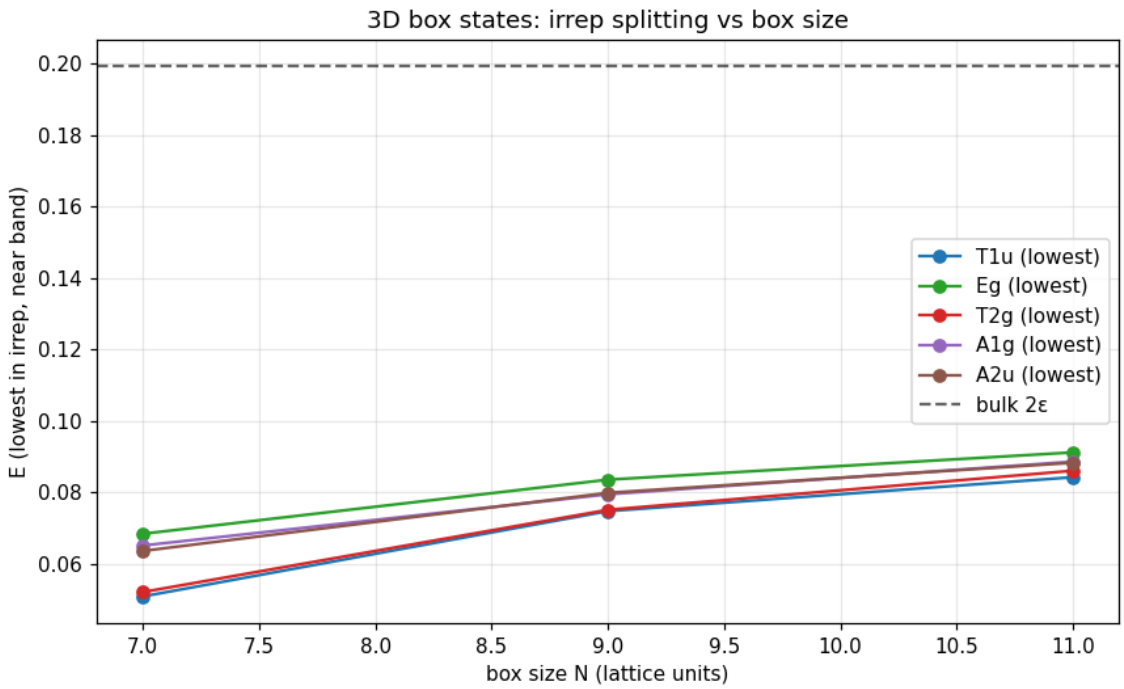

FIG. 6. Lowest energy per O_h irrep as a function of box side N for hard-wall box states on the 3+1D cuboctahedral lattice ($\varepsilon = 0.1$). At every box size T_{1u} is the lowest of the observed band irreps, followed by T_{2g} and E_g. The dashed black line is the bulk band value $E = 2\varepsilon \approx 0.199$. The T_{1u}–E_g gap follows $1/L^2$ to within 2%.

2. In *confinement*, T_{1u} is the lowest-energy band irrep at every box size tested, and other band irreps (T_{2g}, E_g) are separated from it by a gap that scales kinematically as $1/L^2$.

Neither result is forced by any tuning: no mass, coupling, or boundary parameter was adjusted. The effect is purely geometric, arising from the cuboctahedron direction space and the universal $i\varepsilon$ amplitude rule.

Under the continuous rotation group SO(3) the lattice irrep T_{1u} flows to the spin-1 vector representation in the long-wavelength limit $k \to 0$, the same representation carried by every gauge boson in the Standard Model (photon, $W^{\pm}$, Z, gluons). We do not claim to have derived gauge theory from this lattice; we merely note that the geometric ground state of the present model is a vector field, in a sense precisely analogous to how crystal momentum becomes continuous momentum in a long-wavelength limit.

VI. DISCUSSION

A. The exclusion law is universal

The spectral gap (11) depends only on the combinatorics of the N diagonal directions, not on any geometric detail other than "the moves are related by a transitive group action and the amplitude rule is $\delta_{dd'} + i\varepsilon(1 - \delta_{dd'})$". A square lattice with $N = 4$ lightlike directions should therefore give a 3-dimensional physical band with s-wave eigenvalue $1+4i\varepsilon$; this is exactly what Section IV finds for the 2+1D square checkerboard. The law is a geometric fact about the direction space, not about the dimensionality of spacetime.

B. Crystal versus continuous angular momentum

The crystal angular momenta $m \in \{\pm 1, \pm 2, +3\}$ in C_{6v} and the irreps $E_g \oplus T_{2g} \oplus T_{1u} \oplus T_{2u}$ in O_h are discrete: they label finite-group representations, not SO(2) or SO(3) irreps. Like crystal momentum in a solid-state band structure, these labels become continuous in the long-wavelength limit. In that limit T_{1u} flows to a continuous spin-1 vector, T_{2u} to its parity partner, and $E_g \oplus T_{2g}$ to the $\ell = 2$ (d-wave) sector. The $k = 0$ degeneracy of all four irreps within the physical band is therefore a genuine *lattice* signature; it is protected by O_h and broken only away from $k = 0$ or by confinement.

C. Vector ground state—implications

The combined results of Sections IV and V produce a specific and geometric statement: the lowest-energy, freely-propagating content of the equilateral lattice is a vector field. Higher angular-momentum content (E_g, T_{2g},

the d-wave) is not absent—it appears at finite k via the little-group splitting of Fig. 3, and at obstructions via the slit-edge enhancement of Fig. 5—but it is an *excited* response of the lattice, not the ground state.

This is a suggestive geometric counterpart to the empirical fact that every fundamental interaction in nature is mediated by a spin-1 vector boson. It does not, by itself, derive gauge invariance or explain the Standard Model: it merely points out that the *kinematic* vacuum of a geometrically minimal relativistic lattice is a vector field, while scalars and tensors require extra input (obstructions, external momentum, or additional lattice structure).

D. Open questions

The present analysis leaves five questions open. First, the T_{2u} irrep lies outside the 40-eigenvalue shift-invert window in every box size studied. From the band decomposition (9) and the observed ordering (13) we predict $E(T_{2u}) > E(E_g)$, but a direct check requires a wider ARPACK window or an alternative diagonalisation. Second, the band-tracker caveat noted in Ref. [1] remains: at $k > 0$ the 11-fold band fragments into several sub-bands, and identifying the isotropic T_{1u} sub-band requires a more careful eigenvector-overlap tracker than we have implemented here. Third, whether a U(1) or non-abelian gauge symmetry can emerge from couplings between T_{1u} modes is an open question: the T_{1u} ground state carries a 3-component "vector potential" but no gauge constraint has been derived. Fourth, explicit O_h-breaking by non-cubic boxes should shift the level ordering; whether T_{1u} remains the ground state under such perturbations is a natural numerical test. Fifth, far-field interference: a k-dependent physical-band projector would enable Fraunhofer double-slit experiments and direct comparison of the T_{1u} fringe pattern with classical vector-field diffraction theory (see Sec. IV E).

VII. CONCLUSION

We have shown that the internal structure of the equilateral path-integral lattice is fully determined by the geometry of its direction space. The physical propagating band is the traceless part of the N-dimensional diagonal-direction Fourier representation, the s-wave being excluded by a factor-N spectral gap in the local amplitude matrix. In 2+1D this produces a 5-fold band $E_1 \oplus E_2 \oplus B$ carrying crystal angular momenta $m \in \{\pm 1, \pm 2, +3\}$ of C_{6v}. In a natural 3+1D cuboctahedral extension it produces an 11-fold band $E_g \oplus T_{2g} \oplus T_{1u} \oplus T_{2u}$ of the full octahedral group O_h.

Numerical double-slit and hard-wall box experiments on these lattices both single out the vector irrep T_{1u} as the natural low-energy sector: free wave packets are 98% T_{1u}, and the box ground state is T_{1u} at every box size studied, with a T_{1u}–E_g gap that scales as $1/L^2$ to within 2%. Higher-angular-momentum content is only excited at obstructions or at finite momentum.

Geometry → relativistic dispersion [1] → internal structure → vector ground state: the chain from lattice edges to an emergent vector field is now explicit. The natural next step is to study interactions and couplings between T_{1u} modes, and to ask whether gauge invariance can be recovered as a long-wavelength consequence of the lattice geometry alone.

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